Composite Nonlinear Extended State Observer-Based Trajectory Tracking Control for Quadrotor Under Input Constraints

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***Abstract*— In this paper, the trajectory tracking control problem of quadrotor subject to unknown external disturbance and input constraints is addressed. In order to achieve the active disturbance rejection, a composite nonlinear extended state observer (ESO) consisting of both linear and nonlinear functions is developed, and the rigorous convergence analysis is also presented. It is worth mentioning that the proposed composite nonlinear ESO brings together the advantages of both the linear and nonlinear ESOs. Furthermore, the position controller is proposed in the nested saturation control framework, and the disturbance estimate is incorporated in the controller to reject disturbances actively, which provides a composite disturbance rejection control strategy for systems with input constraints and unknown disturbances. The domain of attraction and the rigorous stability analysis of whole system are also presented. Finally, simulation and experiment are provided to show the efficiency of the proposed methods.**

***Index Terms*— Composite nonlinear ESO, nested saturation control, disturbance rejection, trajectory tracking, quadrotor.**

I. INTRODUCTION

IT IS well known that quadrotors have become more and more popular because of its small size, low cost, capability of vertical takeoff and landing (VTOL) and easy to maintain. As one of the most developed one in unmanned aerial vehicles (UAVs), quadrotors have been successfully used in plenty of fields, e.g., surveillance and searching [1], [2], transportation [3], navigation [4], agriculture [5] and construction [6].

The flight control of quadrotor is a comprehensive problem which has attracted great attention, and various classical control strategies have been widely applied in previous works [7], [8], [9], [10], [11]. Note that quadrotors are relatively small in UAVs, which makes them more vulnerable to external disturbances [12]. To deal with this problem, some kinds of advanced control methods, e.g., adaptive control [13], fuzzy logic control [14], H_∞ control [15], disturbance observer based control [16], sliding mode control [17] and finite/fixed time control [18], [19], [20], have been proposed. It should be noted that the availability of the full system states is generally required for the aforementioned methods, however the states might be impossible or expensive to access practically. Recently, the active disturbance rejection control (ADRC) approach proposed in [21] and [22], which has been witnessed as one of the most practical advanced control method, has been applied to quadrotors [23], [24], [25]. The ESO is a key part in ADRC, and by treating the lumped disturbance as an extended state, the ESO is able to estimate both the state and lumped disturbance simultaneously, and the disturbance estimate is incorporated in the controller to achieve active disturbance rejection. However, for the traditional linear ESO (LESO) such as in [26], the initial estimation errors may lead to worse transient performance. A kind of nonlinear ESO (NESO) is proposed in [27] and [28], which avoids the overlarge transient response, but the discontinuous term would brings undesirable chattering phenomenon, which may be a major obstacle for its practical implementation in quadrotors.

Due to the limited thrust of rotors, the constraint on input channel is another non-negligible problem among quadrotors, which may degrade the flight performance or even lead to instability [4], [12], [29]. Besides, the attitude of quadrotors are usually limited in specified range for practical reasons, such as keeping the camera's field of view when combined with visual servoing [6], [30]. In [31], a nested saturation approach is proposed systematically, which has attracted great attention. In [32], a generalized position controller is designed for VTOL vehicles, which considers a class of nested saturation function, and this method is further used in [33] and [34] to deal with state and input constraints. However, the aforementioned methods are established without considering the unknown external disturbances. More recently, some effective disturbance rejection approaches are established for systems with actuator saturation. In [35], a disturbance-observer-based fuzzy model predictive controller has been proposed for attitude control of quadrotor suffering from input constraints and disturbances, but the disturbance estimates are not embedded in the model predictive controller, which may result in the output going beyond the range of constraint. The hyperbolic tangent function is introduced to solve the input saturation

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problem of quadrotor in [36], and a prescribed performance dynamic surface control method is proposed to deal with the model uncertainties and unknown external disturbances, but the design of the saturated controller is also independent of the disturbance estimation, and the boundedness of control input is still not guaranteed. Besides, in [37], the neural network technology has been used to deal with the external disturbances and input saturation in quadrotor, where an event-triggered controller with discrete form is proposed, but the introduction of neural network proposes a higher requirement for the computing ability of hardware.

Motivated by the above literature, this paper addresses the composite disturbance rejection trajectory tracking control problem for quadrotor with input constraints. Compared with the previous works, the disturbance estimation is incorporated into the design process of the saturated controller, which can strictly guarantee the boundedness of control input when the disturbance estimation jumps out of the bound. The main contributions of this paper are summarized as follows.

(i) A novel composite nonlinear ESO (CNESO), possessing the advantages of both the linear and nonlinear ESOs, is developed for estimating the lumped disturbances. It is worth mentioning that, the proposed CNESO not only improves the transient estimation performance effectively, but also guarantees the satisfactory steady-state estimation performance, and the undesired chattering phenomenon in the existing NESOs is eliminated successfully.

(ii) The saturated position controller with active disturbance rejection capability is designed in the nested saturation control framework, which provides an effective strategy for systems with both input constraints and unknown disturbances, and the domain of attraction and the rigorous stability analysis are established.

Numerical simulation and platform experiment are employed to show the effectiveness of the control approaches.

Notations: The symbol $\times$ denotes the cross product operation defined in Section 5.2.2.2 of [38]. The symbols $\lambda_{\min}(\cdot)$ and $\lambda_{\max}(\cdot)$ are the maximum and minimum eigenvalues of a positive definite square matrix, respectively. In addition, $I_3 \in \mathbb{R}^{3\times 3}$ is the identity matrix.

II. System Modeling

In this section, a typical dynamic model of quadrotor is presented. The inertial frame $\mathcal{I} : \{x_I, y_I, z_I\}$ denotes a north-east-down coordinate system fixed to the earth's surface, and the body-fixed frame $\mathcal{B} : \{x_B, y_B, z_B\}$ denotes a coordinate system fixed to the body of quadrotor.

The position and velocity of quadrotor in $\mathcal{B}$ are written as $p^b(t) = [p_x^b(t)\ \ p_y^b(t)\ \ p_z^b(t)]^T$ and $v^b(t) = [v_x^b(t)\ v_y^b(t)\ v_z^b(t)]^T$, respectively. The rotation angles from $\mathcal{I}$ to $\mathcal{B}$ is written as $\eta(t) = [\phi(t)\ \theta(t)\ \psi(t)]^T$, where $\phi(t)$, $\theta(t)$ and $\psi(t)$ denote roll, pitch and yaw angles, respectively. The angular velocity expressed in $\mathcal{B}$ is written as $\omega(t) = [\omega_x(t)\ \omega_y(t)\ \omega_z(t)]^T$. From the Newton-Euler's method, the dynamics of quadrotor expressed in $\mathcal{B}$ is given by [33] and [38]

$$\dot{p}^b(t) = -\omega(t) \times p^b(t) + v^b(t), \tag{1}$$

$$\dot{v}^b(t) = -\omega(t) \times v^b(t) - gR^T(\eta(t))E_3 + \frac{T(t)}{m}E_3 + d_1(t), \tag{2}$$

$$\dot{\eta}(t) = W(\eta(t))\omega(t), \tag{3}$$

$$J\dot{\omega}(t) = -\omega(t) \times J\omega(t) + \tau(t) + G_a(t) + d_2(t), \tag{4}$$

where $E_3 = [0\ 0\ 1]^T$, $T(t)$ and $\tau(t) = [\tau_x(t)\ \tau_y(t)\ \tau_z(t)]^T$ are the inputs about thrust and torques, m is the mass, g is the gravitational acceleration, $J = \mathrm{diag}\{J_x, J_y, J_z\}$ is the inertia matrix and $G_a(t)$ is the gyroscopic torque, $d_1(t)$ and $d_2(t)$ are external disturbances in translational and rotational channels, respectively. The rotation matrices $R(\eta(t))$ and $W(\eta(t))$ are defined the same with that of [33].

Remark 1: For the dynamic model of quadrotor, it is reasonable to assume that: 1) The geometric center is identical to the center of mass; 2) the quadrotor is regarded as a rigid body; 3) the internal mechanical friction is neglected.

Define $p_d(t), v_d(t), a_d(t) \in \mathbb{R}^3$ as the desired position, velocity and acceleration in $\mathcal{I}$, respectively, which satisfies $\dot{p}_d(t) = v_d(t)$, $\dot{v}_d(t) = a_d(t)$. For giving the desired trajectories in $\mathcal{B}$, define $p_d^b(t) = R^T(\eta(t))p_d(t)$, $v_d^b(t) = R^T(\eta(t))v_d(t)$, $a_d^b(t) = R^T(\eta(t))a_d(t)$, then the dynamic equations can be written as

$$\dot{p}_d^b(t) = -\omega(t) \times p_d^b(t) + v_d^b(t),$$
$$\dot{v}_d^b(t) = -\omega(t) \times v_d^b(t) + a_d^b(t).$$

The whole control diagram is shown in Fig. 1, where a hierarchical control structure is adopted here to achieve the desired trajectory tracking. In the positional control loop, the measured position $p(t)$ and desired position $p_d(t)$ in $\mathcal{I}$ are used to calculate the virtual output $u_i(t) = [u_{i1}(t)\ u_{i2}(t)\ u_{i3}(t)]^T$, where a CNESO is adopted to generate the estimations of velocity and lumped disturbance. Let $u_i(t) = -gR^T(\eta_d(t))E_3 + \frac{T(t)}{m}E_3 - a_d^b(t)$ where $\eta_d(t) = [\phi_d(t)\ \ \theta_d(t)\ \ \psi_d(t)]^T$ and $a_d^b(t) = [a_{dx}^b(t)\ a_{dy}^b(t)\ a_{dz}^b(t)]^T$, then the total thrust $T(t)$ and desired angles $\phi_d(t)$ and $\theta_d(t)$ can be derived by the following equations:

$$\theta_d(t) = \arcsin\frac{u_{i1}(t) + a_{dx}^b(t)}{g}, \tag{5}$$

$$\phi_d(t) = -\arcsin\frac{u_{i2}(t) + a_{dy}^b(t)}{g\cos(\theta_d(t))}, \tag{6}$$

$$T(t) = m(u_{i3}(t) + a_{dz}^b(t) + g\cos(\theta_d(t))\cos(\phi_d(t))), \tag{7}$$

where the desired yaw angle $\psi_d(t)$ can be predefined. In the angular control loop, an attitude controller is designed to track the desired angles $\eta_d(t)$ according to the measured angle $\eta(t)$ and angular velocity $\omega(t)$.

Here, we mainly focus on the analysis and design of the positional control loop. First, a simplification of position tracking error system is presented. Define the position and velocity tracking errors $p_e(t) = [p_{ex}(t)\ \ p_{ey}(t)\ \ p_{ez}(t)]^T = p^b(t) - p_d^b(t)$ and $v_e(t) = v^b(t) - v_d^b(t)$, respectively, then the error system can be given by

$$\dot{p}_e(t) = -\omega(t) \times p_e(t) + v_e(t),$$
$$\dot{v}_e(t) = -\omega(t) \times v_e(t) + u_i(t) + d_0(t),$$

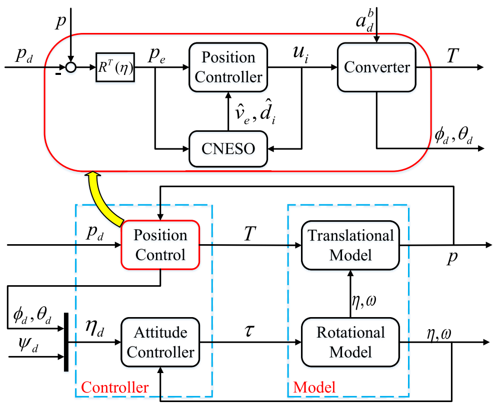

Fig. 1. Control architecture.

where $d_0(t) = gR^T(\eta_d(t))E_3 - gR^T(\eta(t))E_3 + d_1(t)$ includes internal uncertainties and external disturbances. Define a new variable $p_v(t) = [p_{vx}(t)\ p_{vy}(t)\ p_{vz}(t)]^T = -\omega(t) \times p_e(t) + v_e(t)$, we have

$$\dot{p}_e(t) = p_v(t),$$
$$\dot{p}_v(t) = u_i(t) + d_i(t),$$

where $d_i(t) = [d_{ix}(t)\ d_{iy}(t)\ d_{iz}(t)]^T = -\frac{\mathrm{d}(\omega(t)\times p_e(t))}{\mathrm{d}t} - \omega(t) \times v_e(t) + d_0(t)$. For simplicity, only the x–axis channel is analyzed in the following discussions. Take the states as $x_1(t) = p_{ex}(t)$, $x_2(t) = p_{vx}(t)$, $u(t) = u_{i1}(t)$ and $d(x_1, x_2, t) = d_{ix}(t)$, then the error system in x–axis channel is given by

$$\dot{x}_1(t) = x_2(t), \tag{8}$$
$$\dot{x}_2(t) = u(t) + d(x_1, x_2, t). \tag{9}$$

It is worth mentioning that, for system (8)–(9), $x_1(t)$ is always available and can be obtained from the information of position tracking error directly, while $x_2(t)$ is not easy to obtain in absence of velocity sensor. Moreover, $u(t)$ is usually limited, and should be considered during the controller design. In what follows, the ESO will be designed to estimate both $x_2(t)$ and the lumped disturbance $d(x_1, x_2, t)$ in (9), and a saturated position controller with the ability of active disturbance rejection is designed for system (8) and (9) in the following two sections. To this end, an assumption about the external disturbance $d_1(t)$ is given.

Assumption 1: The external disturbance $d_1(t) = [d_{11}(t)\ \ d_{12}(t)\ \ d_{13}(t)]^T$ is a $\mathcal{C}^1$-function and satisfies $|d_{1i}(t)| < u_{mi}$ $(i = 1, 2, 3)$, where u_{mi} is the saturation bound of position controller, and $\dot{d}_1(t)$ is also bounded.

Remark 2: In practice, $d_1(t)$ is mainly composed by wind disturbance, which is continuously differentiable and bounded. Note that the Assumption 1 is to assume that the magnitude of disturbance is within the range of controller, which is reasonable in practice, and the precise upper bound is not needed in this paper.

Remark 3: Generally speaking, the inner attitude control loop possesses faster convergence speed compared with the outer positional control loop in quadrotor control. Thus, it is reasonable to consider that $d_0(t)$ is approximately equal to $d_1(t)$ when designing the outer loop controller. Furthermore, the inner loop controller should be designed to guarantee the stability of whole system. This yields the bounded angular states, which means that both $\omega(t)$ and $\dot{\omega}(t)$ are bounded. All these conditions along with Assumption 1 come to a conclusion that both $d(x_1, x_2, t)$ and $\dot{d}(x_1, x_2, t)$ are continuous differentiable, and would not tend to infinity on finite time.

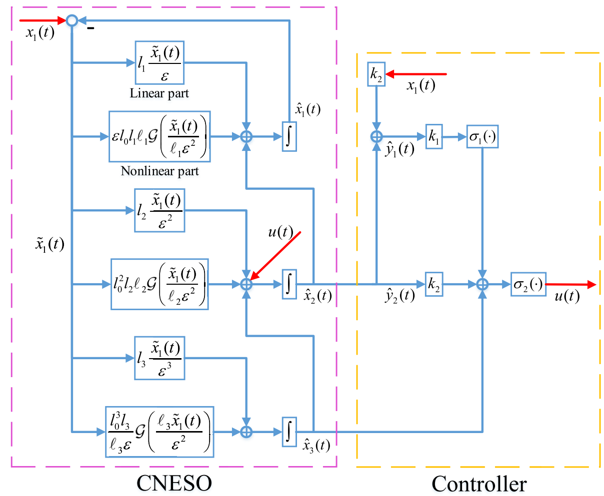

Fig. 2. The structure of CNESO and controller.

III. Design of Composite Nonlinear ESO

Firstly, a CNESO is designed to estimate both the state and disturbance. Let the disturbance $d(x_1, x_2, t)$ in (9) be an extended state $x_3(t)$, and $h(t) = \dot{d}(x_1, x_2, t)$, then the extended system can be given as

$$\dot{x}_1(t) = x_2(t), \tag{10}$$
$$\dot{x}_2(t) = x_3(t) + u(t), \tag{11}$$
$$\dot{x}_3(t) = h(t). \tag{12}$$

Inspired by the LESO (1.3) and NESO (1.4) in [27], the CNESO is designed as

$$\dot{\hat{x}}_1(t) = \hat{x}_2(t) + l_1\left[\frac{\tilde{x}_1(t)}{\varepsilon} + \varepsilon l_0 \ell_1 \mathcal{G}\left(\frac{\tilde{x}_1(t)}{\ell_1 \varepsilon^2}\right)\right], \tag{13}$$
$$\dot{\hat{x}}_2(t) = \hat{x}_3(t) + l_2\left[\frac{\tilde{x}_1(t)}{\varepsilon^2} + l_0^2 \ell_2 \mathcal{G}\left(\frac{\tilde{x}_1(t)}{\ell_2 \varepsilon^2}\right)\right] + u(t), \tag{14}$$
$$\dot{\hat{x}}_3(t) = l_3\left[\frac{\tilde{x}_1(t)}{\varepsilon^3} + \frac{l_0^3}{\ell_3 \varepsilon}\mathcal{G}\left(\ell_3 \cdot \frac{\tilde{x}_1(t)}{\varepsilon^2}\right)\right] \tag{15}$$

where $\tilde{x}_1(t) = x_1(t) - \hat{x}_1(t)$, $\mathcal{G}(s) = \tanh(|s|^{1-\alpha})|s|^{\alpha}\mathrm{sign}(s)$ with $0 < \alpha < 1$, is a nonlinear function borrowed from [39], $l_i > 0$, $i = 1, 2, 3$, $0 < \varepsilon < 1$, and $l_0 > 1$, $\ell_i > 1$, $i = 1, 2, 3$, are observer gains.

The structure of CNESO and controller is shown in Fig. 2.

Remark 4: The parameters l_1, l_2, l_3 and ε are used for tuning the tracking speed and accuracy of CNESO, l_0 represents the proportion of nonlinear part and linear part, and ℓ_1, ℓ_2, ℓ_3, α are used for tuning the nonlinear function. The nonlinear function $\mathcal{G}(s)$ has the following properties: 1) since $|\tanh(s)| < 1$, if $|s|$ is large enough, $\mathcal{G}(s) \approx |s|^{\alpha}\mathrm{sign}(s)$, which is the non-smooth function used in [27] and [28], and can provide better transient performance; 2) when s is near the origin, $\tanh(s) \approx s$, and $\mathcal{G}(s)$ reduces to a linear function, thus the chattering phenomenon is eliminated completely.

Now, we will analyze the convergence of the proposed CNESO. Define the estimation errors as $\tilde{x}_i(t) = x_i(t) - \hat{x}_i(t)$, $i = 1, 2, 3$. According to the above definitions, the estimation error system is

$$\begin{aligned}\dot{\tilde{x}}_1(t) &= \tilde{x}_2(t) - \frac{l_1}{\varepsilon}\tilde{x}_1(t) - l_0 l_1 \ell_1 \varepsilon \mathcal{G}\left(\frac{\tilde{x}_1(t)}{\ell_1 \varepsilon^2}\right),\\ \dot{\tilde{x}}_2(t) &= \tilde{x}_3(t) - \frac{l_2}{\varepsilon^2}\tilde{x}_1(t) - l_0^2 l_2 \ell_2 \mathcal{G}\left(\frac{\tilde{x}_1(t)}{\ell_2 \varepsilon^2}\right),\\ \dot{\tilde{x}}_3(t) &= -\frac{l_3}{\varepsilon^3}\tilde{x}_1(t) - \frac{l_0^3 l_3}{\ell_3 \varepsilon}\mathcal{G}\left(\ell_3 \cdot \frac{\tilde{x}_1(t)}{\varepsilon^2}\right) + h(t).\end{aligned}$$

Then, by applying a transformation of $e_i(t) = \frac{\tilde{x}_i(\varepsilon t)}{\varepsilon^{3-i}}$, $i = 1, 2, 3$, the estimation error system can be rewritten as the following compact form

$$\dot{e}(t) = Ae(t) - G(e_1(t)) + B\varepsilon h(\varepsilon t) \tag{16}$$

where $e(t) = [e_1(t)\ e_2(t)\ e_3(t)]^T$, $B = [0 \quad 0 \quad 1]^T$ and

$$A = \begin{bmatrix} -l_1 & 1 & 0\\ -l_2 & 0 & 1\\ -l_3 & 0 & 0\end{bmatrix},\ G(e_1(t)) = \begin{bmatrix} l_0 l_1 \ell_1 \mathcal{G}\left(\frac{e_1(t)}{\ell_1}\right)\\ l_0^2 l_2 \ell_2 \mathcal{G}\left(\frac{e_1(t)}{\ell_2}\right)\\ \frac{l_0^3 l_3}{\ell_3}\mathcal{G}(\ell_3 e_1(t))\end{bmatrix}.$$

The following two lemmas are necessary for the convergence analysis of estimation error system.

Lemma 1: For any $s \in \mathbb{R}$, $|\mathcal{G}(s)| \le |s|$.

Proof: Define $f_1(s) = s - \tanh(s)$, and it is easy to verify that the inequality $\frac{\partial f_1(s)}{\partial s} = \tanh^2(s) > 0$ holds for $s > 0$. Observing that $f_1(0) = \left.\frac{\partial f_1(s)}{\partial s}\right|_{s=0} = 0$, we have $f_1(s) \ge 0$ for $s \ge 0$. Thus, for any $s \in \mathbb{R}$, $f_1(|s|^{1-\alpha}) \ge 0$ leads to $|s|^{1-\alpha} - \tanh(|s|^{1-\alpha}) \ge 0$, which yields $|\mathcal{G}(s)| = \tanh(|s|^{1-\alpha})|s|^{\alpha} \le |s|$.

Lemma 2: Let $\frac{\partial \mathcal{G}(s)}{\partial s} = \gamma(s)$. For any $s > 0$ and $\ell > 1$, the inequality $\gamma\left(\frac{s}{\ell}\right) - \gamma(s) > 0$ holds.

Proof: Define $f_2(s) = \ell^{1-\alpha}\tanh\left(\frac{s}{\ell^{1-\alpha}}\right) - \tanh(s)$, then we have

$$\frac{\partial f_2(s)}{\partial s} = \tanh^2(s) - \tanh^2\left(\frac{s}{\ell^{1-\alpha}}\right).$$

Note that the tanh function is monotone increasing and for all $s > 0$, $\tanh(s) > 0$. Thus, for any $s > 0$ and $\ell > 1$, the inequality $\tanh^2(s) - \tanh^2\left(\frac{s}{\ell^{1-\alpha}}\right) > 0$ holds, thus $\frac{\partial f_2(s)}{\partial s} > 0$, which together with the fact that $f_2(0) = \left.\frac{\partial f_2(s)}{\partial s}\right|_{s=0} = 0$ yields $f_2(s) > 0$. Note that the derivative of $\mathcal{G}(s)$ along s is

$$\frac{\partial \mathcal{G}(s)}{\partial s} = \alpha|s|^{\alpha-1}\tanh(|s|^{1-\alpha}) + (1-\alpha)(1-\tanh^2(|s|^{1-\alpha})).$$

Then, for any $s > 0$ and $\ell > 1$,

$$\begin{aligned}&\gamma\left(\frac{s}{\ell}\right) - \gamma(s)\\ &= \alpha|s|^{\alpha-1}\left[\ell^{1-\alpha}\tanh\left(\left|\frac{s}{\ell}\right|^{1-\alpha}\right) - \tanh(|s|^{1-\alpha})\right]\\ &\quad + (1-\alpha)\left[\tanh^2(|s|^{1-\alpha}) - \tanh^2\left(\left|\frac{s}{\ell}\right|^{1-\alpha}\right)\right]\\ &= \alpha|s|^{\alpha-1} f_2(|s|^{1-\alpha}) + (1-\alpha)\frac{\partial f_2(|s|^{1-\alpha})}{\partial |s|^{1-\alpha}}\\ &> 0,\end{aligned}$$

which completes the proof.

Consider the Lyapunov function given by

$$V(e(t)) = e^T(t)Pe(t) + \lambda\int_0^{e_1(t)}\mathcal{G}(s)\mathrm{d}s,$$

where $\lambda = \frac{l_2^2 + l_3^2 + l_1 l_3 + l_2}{l_1 l_2 - l_3}$, P is the positive definite symmetric solution of Lyapunov equation $PA + A^T P = -I_3$, and can be determined as

$$P = \begin{bmatrix} p_{11} & p_{12} & p_{13}\\ p_{12} & p_{22} & p_{23}\\ p_{13} & p_{23} & p_{33}\end{bmatrix},$$

where $p_{11} = \frac{l_2^2 + l_3^2 + l_1 l_3 + l_2}{2(l_1 l_2 - l_3)}$, $p_{22} = -p_{13} = \frac{l_1^2 + l_1 l_3 + l_2 + 1}{2(l_1 l_2 - l_3)}$, $p_{12} = p_{23} = -\frac{1}{2}$, $p_{33} = \frac{l_1^3 + l_1^2 l_3 + l_2^2 l_1 - l_2 l_3 + l_1 + l_3}{2 l_3 (l_1 l_2 - l_3)}$.

Define $\lambda_1 = \lambda_{\min}(P)$, $\lambda_2 = \lambda_{\max}(P) + \frac{1}{2}\lambda$ and $0 < \beta < \frac{1}{2}$. The following theorem establishes the convergence conditions of the estimation error system.

Theorem 1: Consider the estimation error system (8)–(9). If the observer gains satisfy $l_1 l_2 - l_3 > 0$, and $|h(t)| < h_m$, where h_m is a positive constant, then there exist $l^* > 1$, $\ell^* > 1$ such that for any $l_0 \in [1, l^*]$ and $\ell_i \in [1, \ell^*]$, $i = 1, 2, 3$, the estimation errors are uniformly ultimately bounded (UUB), and satisfy

$$\begin{aligned}|\tilde{x}_i(t)| \le \varepsilon^{3-i}&\left[\sqrt{\frac{V(e(0))}{\lambda_1}} - \frac{2\varepsilon\|PB\|h_m\lambda_2}{\beta\lambda_1}\right]\mathrm{e}^{-\frac{\beta t}{2\lambda_2\varepsilon}}\\ &+ \varepsilon^{4-i}\frac{2\|PB\|h_m\lambda_2}{\beta\lambda_1},\ \forall t \ge 0, i = 1, 2, 3.\end{aligned} \tag{17}$$

Proof: From (16), we obtain

$$\begin{aligned}\dot{e}(t) &= Ae(t) - L\mathcal{G}(e_1(t)) - (l_0 - 1)L'\mathcal{G}(e_1(t))\\ &\quad + F_0(l_0, \ell, e_1(t)) + B\varepsilon h(\varepsilon t)\\ &= Ae(t) - L\mathcal{G}(e_1(t)) + F(l_0, \ell, e_1(t)) + B\varepsilon h(\varepsilon t),\end{aligned}$$

where

$$\begin{aligned}L &= [l_1,\ l_2,\ l_3]^T,\\ L' &= [l_1,\ l_2(l_0 + 1),\ l_3(l_0^2 + l_0 + 1)]^T,\\ F_0(l_0, \ell, e_1(t)) &= \begin{bmatrix} -l_0 l_1\left(\ell_1\mathcal{G}\left(\frac{e_1(t)}{\ell_1}\right) - \mathcal{G}(e_1(t))\right)\\ -l_0^2 l_2\left(\ell_2\mathcal{G}\left(\frac{e_1(t)}{\ell_2}\right) - \mathcal{G}(e_1(t))\right)\\ \frac{l_0^3 l_3}{\ell_3}\left(\ell_3\mathcal{G}(e_1(t)) - \mathcal{G}(\ell_3 e_1(t))\right)\end{bmatrix},\\ F(l_0, \ell, e_1(t)) &= -(l_0 - 1)L'\mathcal{G}(e_1(t)) + F_0(l_0, \ell, e_1(t)).\end{aligned}$$

Define $F_1(\ell, e_1(t)) = \ell\mathcal{G}\left(\frac{e_1(t)}{\ell}\right) - \mathcal{G}(e_1(t))$, and denote the partial derivative of $F_1(\ell, e_1(t))$ w.r.t. $e_1(t)$ by $F_1'(\ell, e_1(t))$. It follows from Lemma 2 that for any $e_1(t) > 0$ and $\ell > 1$,

$$\begin{aligned}F_1'(\ell, e_1(t)) &= \frac{\partial F_1(\ell, e_1(t))}{\partial e_1(t)}\\ &= \gamma\left(\frac{e_1(t)}{\ell}\right) - \gamma(e_1(t)) > 0,\end{aligned}$$

$$\left.F_1'(\ell, e_1(t))\right|_{e_1(t)=0} = 0,\quad \lim_{e_1(t)\to+\infty} F_1'(\ell, e_1(t)) = 0.$$

Note that $F_1'(\ell, e_1(t))$ is a continuous even function about $e_1(t)$, and it is clear that $F_1'(\ell, e_1(t))$ is bounded for fixed ℓ. Assume the upper bound of $F_1'(\ell, e_1(t))$ is $\delta(\ell)$. According to the Lagrange mean value theorem, for any $s_1, s_3 \in \mathbb{R}$ with $s_1 < s_3$, there exists $s_2 \in \mathbb{R}$ satisfying $s_1 < s_2 < s_3$ and $F_1(\ell, s_3) - F_1(\ell, s_1) = \frac{\partial F_1(\ell, s_2)}{\partial s_2}(s_3 - s_1)$. Thus,

$$\begin{aligned}|F_1(\ell, s_3) - F_1(\ell, s_1)| &= \left|\frac{\partial F_1(\ell, s_2)}{\partial s_2}\right| |s_3 - s_1| \\ &\leq \delta(\ell)|s_3 - s_1|,\end{aligned}$$

which means that $F_1(\ell, e_1(t))$ is globally Lipschitz with a Lipschitz constant $\delta(\ell)$. Based on this conclusion and the equation $F_1(\ell, 0) = 0$, the inequality

$$|F_1(\ell, e_1(t))| \leq \delta(\ell)|e_1(t)|$$

is derived. From the definition of $F_0(l_0, \ell, e_1(t))$, we get

$$\begin{aligned}\|F_0(l_0, \ell, e_1(t))\| &= \left\| \begin{bmatrix} -l_0 l_1 F_1(\ell_1, e_1(t)) \\ -l_0^2 l_2 F_1(\ell_2, e_1(t)) \\ \frac{l_0^3 l_3}{\ell_3} F_1(\ell_3, \ell_3 e_1(t)) \end{bmatrix} \right\| \\ &\leq \Delta(l_0, \ell)\|e(t)\|, \\ \Delta(l_0, \ell) &= \sqrt{l_0^2 l_1^2 \delta^2(\ell_1) + l_0^4 l_2^2 \delta^2(\ell_2) + l_0^6 l_3^2 \delta^2(\ell_3)}.\end{aligned}$$

If $\ell_i \to 1$, $i = 1, 2, 3$, then $F_1'(\ell_i, e_1(t)) \to 0$, which leads to $\delta(\ell_i) \to 0$ and $\Delta(l_0, \ell) \to 0$. According to Lemma 1, we get

$$\begin{aligned}\|F(l_0, \ell, e_1(t))\| &= \left\| -(l_0 - 1)L'\mathcal{G}(e_1(t)) + F_0(l_0, \ell, e_1(t)) \right\| \\ &\leq (l_0 - 1)\|L'\| \cdot |e_1(t)| + \Delta(l_0, \ell)\|e(t)\| \\ &\leq \Delta'(l_0, \ell)\|e(t)\|\end{aligned}$$

where $\Delta'(l_0, \ell) = (l_0 - 1)\|L'\| + \Delta(l_0, \ell)$. It can be derived easily that $\Delta'(l_0, \ell) \to 0$ if $l_0 \to 1$ and $\ell_i \to 1$, $i = 1, 2, 3$. Thus, there exist $l^* > 1$ and $\ell^* > 1$ such that for any $l_0 \in [1, l^*]$ and $\ell_i \in [1, \ell^*]$, $i = 1, 2, 3$, the inequality

$$\Delta'(l_0, \ell) \leq \frac{1 - 2\beta}{4\|P\|}$$

holds, where $0 < \beta < \frac{1}{2}$ is a design parameter.

According to Lemma 1 and the definition of $V(e(t))$, we have

$$\begin{aligned}V(e(t)) &\geq e^T(t)Pe(t) \geq \lambda_{\min}(P)\|e(t)\|^2 = \lambda_1\|e(t)\|^2, \\ V(e(t)) &\leq e^T(t)Pe(t) + \lambda \int_0^{e_1(t)} s\mathrm{d}s \\ &\leq \lambda_{\max}(P)\|e(t)\|^2 + \frac{1}{2}\lambda e_1^2(t) \\ &\leq \lambda_2\|e(t)\|^2.\end{aligned}$$

Under the condition $l_1 l_2 - l_3 > 0$, $l_0 \in [1, l^*]$ and $\ell_i \in [1, \ell^*]$, $i = 1, 2, 3$, the derivative of $V(e(t))$ is given by

$$\begin{aligned}\dot{V}(e(t)) &= \dot{e}^T(t)Pe(t) + e^T(t)P\dot{e}(t) + \lambda\mathcal{G}(e_1(t))\dot{e}_1(t) \\ &= -\|e(t)\|^2 - 2e^T(t)PL\mathcal{G}(e_1(t)) \\ &\quad + 2e^T(t)PF(l_0, \ell, e_1(t)) + 2e^T(t)PB\varepsilon h(\varepsilon t) \\ &\quad + \lambda\mathcal{G}(e_1(t))\left[-l_1 e_1(t) + e_2(t) - l_0 l_1 \ell_1 \mathcal{G}\left(\frac{e_1(t)}{\ell_1}\right)\right] \\ &= -\|e(t)\|^2 - (1 + \lambda l_1)e_1(t)\mathcal{G}(e_1(t)) \\ &\quad - \lambda l_0 l_1 \ell_1 \mathcal{G}(e_1(t))\mathcal{G}\left(\frac{e_1(t)}{\ell_1}\right) + e_3(t)\mathcal{G}(e_1(t)) \\ &\quad + 2e^T(t)PF(l_0, \ell, e_1(t)) + 2e^T(t)PB\varepsilon h(\varepsilon t) \\ &\leq -\|e(t)\|^2 + |e_1(t)| \cdot |e_3(t)| \\ &\quad + 2\Delta'(l_0, \ell)\|P\| \cdot \|e(t)\|^2 \\ &\quad + 2\varepsilon\|PB\| \cdot \|e(t)\||h(\varepsilon t)| \\ &\leq -\frac{1}{2}e_1^2(t) - e_2^2(t) - \frac{1}{2}e_3^2(t) - \frac{1}{2}(|e_1(t)| - |e_3(t)|)^2 \\ &\quad + \left(\frac{1}{2} - \beta\right)\|e(t)\|^2 + 2\varepsilon\|PB\|h_m\|e(t)\| \\ &\leq -\beta\|e(t)\|^2 + 2\varepsilon\|PB\|h_m\|e(t)\| \\ &\leq -\frac{\beta}{\lambda_2}V(e(t)) + \frac{2\varepsilon\|PB\|h_m}{\sqrt{\lambda_1}}\sqrt{V(e(t))}.\end{aligned}$$

Since $\dot{V}(e(t)) = 2\sqrt{V(e(t))} \cdot \frac{\partial\sqrt{V(e(t))}}{\partial t}$, it is easy to obtain

$$\frac{\partial\sqrt{V(e(t))}}{\partial t} \leq -\frac{\beta}{2\lambda_2}\sqrt{V(e(t))} + \frac{\varepsilon\|PB\|h_m}{\sqrt{\lambda_1}}, \tag{18}$$

and it can be derived from (18) that

$$\begin{aligned}\|e(t)\| &\leq \sqrt{\frac{V(e(t))}{\lambda_1}} \\ &\leq \sqrt{\frac{V(e(0))}{\lambda_1}}\mathrm{e}^{-\frac{\beta t}{2\lambda_2}} + \frac{\varepsilon\|PB\|h_m}{\lambda_1}\int_0^t \mathrm{e}^{-\frac{\beta(t-s)}{2\lambda_2}}\mathrm{d}s \\ &= \left[\sqrt{\frac{V(e(0))}{\lambda_1}} - \frac{2\varepsilon\|PB\|h_m\lambda_2}{\beta\lambda_1}\right]\mathrm{e}^{-\frac{\beta t}{2\lambda_2}} \\ &\quad + \frac{2\varepsilon\|PB\|h_m\lambda_2}{\beta\lambda_1}.\end{aligned} \tag{19}$$

By combining $|\tilde{x}_i(t)| = \varepsilon^{3-i}|e_i(t/\epsilon)|$, $i = 1, 2, 3$ and (19) together, and according to [28] and [40], we get (17) immediately, which completes the proof. The parameter tuning process of CNESO is given as follows.

Step 1: Removing the nonlinear part in (13)–(15), and tuning ε, l_1, l_2, l_3 by applying the LESO tuning method to get the desired steady-state tracking performance, e.g., $|\tilde{x}_i(t)| \leq \tilde{d}_m$. Then let $l_{10} = l_1$, $l_{20} = l_2$, $l_{30} = l_3$.

Step 2: Adding the nonlinear part, and setting $l_0 = \ell_1 = \ell_2 = \ell_3 = 1$ initially.

Step 3: Increasing l_0 and changing l_1, l_2, l_3 such that $l_0 l_1 = l_{10}$, $l_0^2 l_2 = l_{20}$, $l_0^3 l_3 = l_{30}$ hold.

Step 4: Increasing ℓ_1, ℓ_2 and ℓ_3 to get a better transient performance.

Step 5: Repeating Step 3 and Step 4 until a satisfied transient performance is obtained.

Define $\tilde{x}_m(s) = \max_{t\in[s,\infty),i=1,2,3}|\tilde{x}_i(t)|$. The result of Theorem 1 leads to the following corollary.

Corollary 1: There exists $0 < \varepsilon^* < 1$ such that for any $\varepsilon \in (0, \varepsilon^*)$ and $\xi > 0$, the estimation error satisfies $\tilde{x}_m(t_\varepsilon) \leq \xi$, $\forall t \in [t_\varepsilon, \infty)$, where $t_\varepsilon = \frac{-8\lambda_2\varepsilon\ln\varepsilon}{\beta}$.

Proof: It is easy to verify that, for any $t \geq t_\varepsilon$,

$$|\tilde{x}_i(t)| \leq \varepsilon^{4-i}\Gamma(\varepsilon)$$

holds, where $\Gamma(\varepsilon) = \varepsilon^3\left(\sqrt{\frac{V(e(0))}{\lambda_1}} + \frac{2\varepsilon\|PB\|h_m\lambda_2}{\beta\lambda_1}\right)$, and notice that $\lim_{\varepsilon\to 0} t_\varepsilon = \lim_{\varepsilon\to 0}\Gamma(\varepsilon) = 0$, we have that for any $t_\varepsilon > 0$, $|\tilde{x}_i(t)| \to 0$ uniformly in $t \in [t_\varepsilon, \infty)$ as $\varepsilon \to 0$, which completes the proof.

It should be noted that Theorem 1 and Corollary 1 holds only when $h(t)$ is bounded. However, the limitation of $h(t)$ means that the trajectories of system must be bounded a priori, which may cause the problem of circular reasoning. To avoid this problem, a set-based proof of stability will be given in the next section, and the result of Corollary 1 will be applied on finite time period, which is more rigorous.

IV. Design of the Saturated Controller

In this section, a saturated controller is designed using system output $x_1(t)$ and estimated states $\hat{x}_i(t)$. Let $y_1(t) = k_2x_1(t) + x_2(t)$ and $y_2(t) = x_2(t)$, where $k_2 > 0$ is a design parameter, then system (8)–(9) can be rewritten as

$$\dot{y}_1(t) = k_2y_2(t) + u(t) + d(t), \tag{20}$$

$$\dot{y}_2(t) = u(t) + d(t). \tag{21}$$

Let $\hat{y}_1(t) = k_2x_1(t)+\hat{x}_2(t)$ and $\hat{y}_2(t) = \hat{x}_2(t)$ be the estimates of $y_1(t)$ and $y_2(t)$, respectively, and the estimation errors are defined as $\tilde{y}_1(t) = y_1(t) - \hat{y}_1(t) = \tilde{x}_2(t)$ and $\tilde{y}_2(t) = y_2(t) - \hat{y}_2(t) = \tilde{x}_2(t)$, respectively. Motivated by [31], the saturated controller is then designed as

$$u(t) = -\sigma_2\left(k_2\hat{y}_2(t) + \sigma_1\left(k_1\hat{y}_1(t)\right) + \hat{x}_3(t)\right) \tag{22}$$

where $k_1 > 0$ is a design parameter, $\sigma_1(\cdot)$ and $\sigma_2(\cdot)$ are the saturated functions and satisfy the following properties: 1) σ_i $(i = 1, 2)$ is a continuous and nondecreasing function; 2) $x\sigma_i(x) > 0$ for any $x \neq 0$; 3) $\sigma_i(x) = x$ for any $|x| \leq L_i$; 4) $|\sigma_i(x)| \leq M_i$ for any $x \in \mathbb{R}$; 5) $0 < L_i \leq M_i$ and $M_1 \leq bL_2$ where $0 < b < 1$. Note that $M_2 = u_{m1}$ is the saturation bound of the controller.

For proving the stability of whole system, the following two lemmas will be presented at first.

Lemma 3: Suppose that the function σ_2 in controller (22) enters in its linear region in $t \in [t_{\sigma1}, \infty)$, where $t_{\sigma1} > 0$ is an ε-independent constant, and the conditions of Theorem 1 hold, then there always exist $\varepsilon_1 > 0$ and k_1 such that for all $\varepsilon \in (0, \varepsilon_1)$, the state $y_1(t)$ converges into a set $Q_1 = \{y_1(t) \in \mathbb{R} \mid |k_1y_1(t)| \leq a_1L_1\}$, where a_1 is a tunable parameter [33], and then the saturation function σ_1 works in its linear region.

Proof: When the function σ_2 works in its linear region for $t \in [t_{\sigma1}, \infty)$, the system (20) is equivalent to

$$\dot{y}_1(t) = k_2\tilde{y}_2(t) + \tilde{x}_3(t) - \sigma_1(k_1\hat{y}_1(t)). \tag{23}$$

Define a Lyapunov function $V_1(t) = \frac{1}{2}y_1^2(t)$, and its derivative reads

$$\dot{V}_1(t) = -y_1(t)\sigma_1(k_1\hat{y}_1(t)) + y_1(t)(k_2\tilde{y}_2(t) + \tilde{x}_3(t)).$$

The convergence of $y_1(t)$ can be guaranteed by $\dot{V}_1(t) < 0$ for all $y_1(t) \notin Q_1$, which means $|k_1y_1(t)| > a_1L_1$, then

$$|k_1\hat{y}_1(t)| = |k_1y_1(t) - k_1\tilde{y}_1(t)| > a_1L_1 - k_1\tilde{d}_{m1},$$

where $\tilde{d}_{m1} = \tilde{x}_m(t_{\sigma1})$. If $a_1L_1 - k_1\tilde{d}_{m1} > 0$, then the sign of $k_1\hat{y}_1(t)$ is determined by that of $k_1y_1(t)$, which implies

$$\begin{aligned}\dot{V}_1(t) &= -|y_1(t)| \cdot \left|\sigma_1(k_1\hat{y}_1(t))\right| + y_1(t)(k_2\tilde{y}_2(t) + \tilde{x}_3(t)) \\ &\leq -(a_1L_1 - k_1\tilde{d}_{m1})|y_1(t)| + (k_2 + 1)\tilde{d}_{m1}|y_1(t)| \\ &< -\frac{a_1L_1}{k_1}\left[a_1L_1 - (k_1 + k_2 + 1)\,\tilde{d}_{m1}\right].\end{aligned}$$

The condition $\dot{V}_1(t) < 0$ is satisfied if

$$a_1L_1 - (k_1 + k_2 + 1)\,\tilde{d}_{m1} > 0 \tag{24}$$

holds, then $y_1(t)$ will converge into Q_1. Once $y_1(t)$ enters in Q_1, we have

$$|k_1\hat{y}_1(t)| = |k_1y_1(t) - k_1\tilde{y}_1(t)| \leq a_1L_1 + k_1\tilde{d}_{m1}.$$

When the condition

$$(a_1 - 1)L_1 + k_1\tilde{d}_{m1} \leq 0 \tag{25}$$

holds, $|k_1\hat{y}_1(t)| \leq L_1$, and σ_1 works in its linear region.

From the above discussions, we can conclude that the inequalities (24) and (25) are the necessary conditions for the conclusion of Lemma 3, which yields $(k_1 + k_2 + 1)\tilde{d}_{m1} < a_1L_1 \leq L_1 - k_1\tilde{d}_{m1}$. If a_1 exists, the inequality $L_1 > (2k_1 + k_2 + 1)\tilde{d}_{m1}$ should be satisfied. Thus, the inequality

$$L_1 > (k_2 + 1)\tilde{d}_{m1} \tag{26}$$

holds for the existence of k_1. When k_2 is fixed, we can always find a $\tilde{d}^*_{m1} > 0$ satisfying (26) for all $\tilde{d}_{m1} \in (0, \tilde{d}^*_{m1})$. According to the conclusions of Theorem 1 and Corollary 1, there exists an $\varepsilon_1 > 0$ such that $\tilde{d}_{m1} \in (0, \tilde{d}^*_{m1})$ hold for all $\varepsilon \in (0, \varepsilon_1)$, which completes the proof.

Lemma 4: Suppose that the functions σ_1 and σ_2 enter in their linear region for $t \in [t_{\sigma2}, \infty)$, respectively, where $t_{\sigma2} > 0$ is an ε-independent constant, and the conditions of Theorem 1 hold, then there always exist $\varepsilon_2 > 0$ such that for all $\varepsilon \in (0, \varepsilon_2)$, the states of the error system (8)–(9) are UUB, and exponentially convergent into the region $\chi = \{x(t) = [x_1(t)\ x_2(t)]^T \in \mathbb{R}^2 \big| \, |x_1(t)| \leq x_{1m},\ |x_2(t)| \leq x_{2m}\}$, where $x_{1m} = \frac{k_1+k_2}{k_1k_2(k_2-k_1)}f_m$, $x_{2m} = \frac{2f_m}{k_2-k_1}$ are positive constants.

Proof: When σ_1 and σ_2 work in their linear region for $t \in [t_{\sigma2}, \infty)$, the saturated control input becomes $u(t) = -k_1\hat{y}_1(t) - k_2\hat{y}_2(t) - \hat{x}_3(t)$. Substituting it into (20)–(21) yields

$$\dot{y}_1(t) = -k_1y_1(t) + \tilde{f}(t), \tag{27}$$

$$\dot{y}_2(t) = -k_1y_1(t) - k_2y_2(t) + \tilde{f}(t) \tag{28}$$

where $\tilde{f}(t) = k_1\tilde{y}_1(t) + k_2\tilde{y}_2(t) + \tilde{x}_3(t)$. The upper bound of $\tilde{f}(t)$ is $|\tilde{f}(t)| \leq (k_1+k_2+1)\tilde{d}_{m2} = f_m$, where $\tilde{d}_{m2} = \tilde{x}_m(t_{\sigma2})$. According to Theorem 1 and Corollary 1, there exists an $\varepsilon_2 > 0$ such that $\tilde{d}_{m2} \in (0, \tilde{d}^*_{m2})$ hold for all $\varepsilon \in (0, \varepsilon_2)$, where $\tilde{d}^*_{m2} > 0$ is a design parameter. According to the definition of $y_1(t)$ and $y_2(t)$, for any $k_2 > k_1 > 0$, it is easy to verify that the states $x_1(t)$ and $x_2(t)$ are bounded by

$$\begin{aligned}|x_1(t)| \leq{}& \frac{|y_1(t_{\sigma2})|}{k_2 - k_1}\mathrm{e}^{-k_1(t-t_{\sigma2})} + \frac{|y_2(t_{\sigma2})|}{k_2 - k_1}\mathrm{e}^{-k_2(t-t_{\sigma2})} \\ &+ \frac{k_1 + k_2}{k_1k_2(k_2 - k_1)}f_m, \\ |x_2(t)| \leq{}& \frac{k_1|y_1(t_{\sigma2})|}{k_2 - k_1}\mathrm{e}^{-k_1(t-t_{\sigma2})} + \frac{k_2|y_2(t_{\sigma2})|}{k_2 - k_1}\mathrm{e}^{-k_2(t-t_{\sigma2})} \\ &+ \frac{2f_m}{k_2 - k_1}.\end{aligned}$$

The ultimate bounds of states are $\lim_{t\to\infty}|x_1(t)| \leq \frac{k_1+k_2}{k_1k_2(k_2-k_1)}f_m$ and $\lim_{t\to\infty}|x_2(t)| \leq \frac{2f_m}{k_2-k_1}$, which completes the proof.

According to the Assumption 1 and Remark 3, the lumped disturbance is a $\mathcal{C}^1$-function, then for any $x(t) = [x_1(t)\ x_2(t)]^T$ belonging to the domain $\{x(t) \in \mathbb{R}^2 \mid \|x(t)\| \leq x_m\}$, where $x_m > 0$ is a design parameter, there exist two constants $C_1 > 0$ and $C_2 \in (0, M_2)$ such that $|d(x_1, x_2, t)| \leq C_1\|x(t)\| + C_2$. Define two sets $\Omega_1 = \{x(t) \in \mathbb{R} \mid \|x(t)\| \leq \rho - \kappa\}$, and $\Omega_2 = \{x(t) \in \mathbb{R} \mid \|x(t)\| \leq \rho - \frac{\kappa}{2}\}$, where $\rho = \min\{x_m, \frac{M_2-C_2}{C_1}\}$, $\kappa \in (0, \rho)$ [41]. For proving the stability of whole system, the following proposition is presented.

Proposition 1: Consider the system (8)–(9) and the saturated controller (22), if the Assumption 1 holds and the initial state $x(0)$ is the interior point of Ω_1, then there exists an $\varepsilon_0 > 0$ such that for any $\varepsilon \in (0, \varepsilon_0)$ in CNESO (13)–(15) and $\kappa \in (0, \rho)$, the state $x(t) \in \Omega_2$, $\forall t \in [0, \infty)$.

Proof: It is easily derived from the equation (8)-(9) that when $x(t) \in \Omega_2$, the following inequality

$$\|x(t)\| \leq \left\|e^{A_0 t}x(0)\right\| + \left\|\int_0^t e^{A_0(t-s)}B_0(M_2 + C_1\|x(s)\| + C_2)\mathrm{d}s\right\|$$

holds, where $A_0 = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ and $B_0 = [0,\ 1]^T$. Note that the trajectory of system (8)-(9) is independent of ε, then there exists an ε-independent time $t_0 > 0$ satisfying that $x(t) \in \Omega_1$, $\forall t \in [0, t_0]$. Suppose that the conclusion in Proposition 1 is false, then for any $\varepsilon > 0$, there exist t_1 and t_2 satisfying $t_2 > t_1 > t_0$ and

$$\begin{cases} x(t_1) \in \partial\Omega_1,\ x(t_2) \in \partial\Omega_2, \\ \{x(t) \mid t \in (t_1, t_2]\} \subset \Omega_2 - \Omega_1, \\ \{x(t) \mid t \in [0, t_2]\} \subset \Omega_2, \end{cases} \tag{29}$$

where $\partial\Omega_1$, $\partial\Omega_2$ are the sets of boundary points of Ω_1 and Ω_2 [42]. According to the Remark 3, there exist two ε-independent constants d_m^* and h_m^* satisfying $|d(x_1, x_2, t)| \leq d_m^*$ and $|h(t)| \leq h_m^*$ for any $t \in [0, t_2]$. It can be seen easily from the above definition that $M_2 > d_m^*$, then there exist $\varepsilon_3 > 0$, $b > 0$ and $\rho > 0$, such that for any $\varepsilon \in (0, \varepsilon_3)$ and $t \in [t_1, t_2]$, the inequalities

$$|\tilde{x}_i(t)| \leq \tilde{d}_m^* = \frac{M_2 - d_m^* - k_2\rho - bM_2}{k_2 + 1},\ i = 1, 2, 3 \tag{30}$$

hold according to Corollary 1. When $\varepsilon \in (0, \varepsilon_3)$ and $t \in [t_1, t_2]$, it is derived that

$$\begin{aligned} |k_2\hat{y}_2(t) + \sigma_1(k_1\hat{y}_1(t)) + \hat{x}_3(t)| &\leq (k_2+1)\tilde{d}_m^* + d_m^* + k_2\rho + bM_2 \\ &\leq M_2 \end{aligned} \tag{31}$$

which means the $\sigma_2(\cdot)$ in (22) works in its linear region. According to Lemma 3, there exist $\varepsilon_0 = \min\{\varepsilon_1, \varepsilon_3\}$ and k_1 such that for all $\varepsilon \in (0, \varepsilon_0)$, the state $y_1(t) = k_2x_1(t) + x_2(t)$ converges into Q_1, and the convergence time is represented as t_Q. In what follows, three cases will be analyzed for $t \in [t_1, t_2]$, respectively.

Case 1: When $t_Q \leq t_1$, which means that $y_1(t) \in Q_1$, $\forall t \in [t_1, t_2]$, the saturation function σ_1 would work in its linear region. According to Lemma 4, $x_1(t)$ and $x_2(t)$ would exponentially convergent to the region χ in $t \in [t_1, t_2]$ if k_1 and k_2 are large enough. It also means $\|x(t_1)\| > \|x(t_2)\|$, which contradicts with (29).

Case 2: When $t_Q > t_2$, which means that $y_1(t) \notin Q_1$, $\forall t \in [t_1, t_2]$, denote the Lyapunov function $V_t(x(t)) = V_1(t) = \frac{1}{2}(k_2x_1(t) + x_2(t))^2$, then for all $\varepsilon \in (0, \varepsilon_0)$, $\dot{V}_t(x(t)) < 0$ holds from the derivation of Lemma 3. It can be easily derived that the trajectories of system would tend to the origin, and then $\|x(t_1)\| > \|x(t_2)\|$, which also contradicts with (29).

Case 3: When $t_1 < t_Q \leq t_2$, the analysis can be separated into two time periods $t \in (t_1, t_Q]$ and $t \in (t_Q, t_2]$. Then, by applying the conclusion of the above two cases, $\|x(t_1)\| > \|x(t_Q)\| > \|x(t_2)\|$ is derived, which contradicts with (29).

Thus, Proposition 1 is proved.

From the above derivations, we can derive the stability of system (8)–(9) for given CNESO and saturated controller.

Theorem 2: Consider the error system (8)–(9), CNESO (13)–(15) and the saturated controller (22). If the Assumption 1 holds and the initial state $x(0)$ is the interior point of Ω_1, then there exist positive constants ε_0, k_1, k_2 and b such that for any $t > 0$, $\varepsilon \in (0, \varepsilon_0)$ and $\kappa \in (0, \rho)$, the states of the error system are uniformly ultimately bounded (UUB), and exponentially convergent into a bounded region.

Proof: From the conclusion of Proposition 1 and the continuity of $d(x_1, x_2, t)$ and $h(t)$, there exist two ε-independent constant $\bar{d}_m$ and $\bar{h}_m$ such that $|d(x_1, x_2, t)| \leq \bar{d}_m \leq M_2$ and $h(t) \leq \bar{h}_m$ hold for any $t > 0$, $\varepsilon \in (0, \varepsilon_0)$ and $\kappa \in (0, \rho)$. According to the analysis of Lemma 3 and Lemma 4, the states of system (8)–(9) are UUB, and exponentially convergent into the bounded region χ, and the proof is completed.

V. Numerical Simulations

In this section, numerical examples are provided to illustrate the validity of the proposed control approaches.

A. Comparative Results Among Different ESOs

In this subsection, the comparative study is carried out among the existing LESO, NESO and the proposed CNESO (13)–(15). Specially, the traditional LESO and NESO are borrowed from [27], respectively, and the parameter ε is set as 0.025.

In the comparative study, a simple ADRC controller

$$u(t) = -k_1y(t) - k_2\hat{x}_2(t) - \hat{x}_3(t)$$

is applied to a second-order system

$$\begin{aligned} \dot{x}_1(t) &= x_2(t) \\ \dot{x}_2(t) &= u(t) + d(t), \end{aligned}$$

where the disturbance (unit: N) is assumed to be

$$d(t) = \begin{cases} \mathrm{sign}(\sin(\pi t)), & t < 1.5\mathrm{s} \\ \sin(\pi t), & t > 1.5\mathrm{s} \end{cases}$$

the parameters of controller are set to be $k_1 = 1$ and $k_2 = 2$, and the parameters of CNESO are set to be $\varepsilon = 0.007$, $l_0 = 5$, $l_1 = 0.6$, $l_2 = 0.12$, $l_3 = 0.008$, $\ell_1 = 7500$, $\ell_2 = 50$,

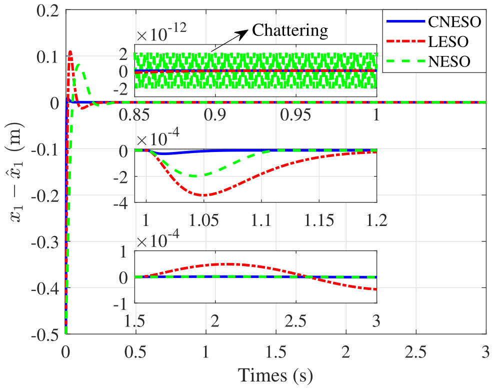

Fig. 3. The estimation error signals of $x_1(t)$ for different ESOs.

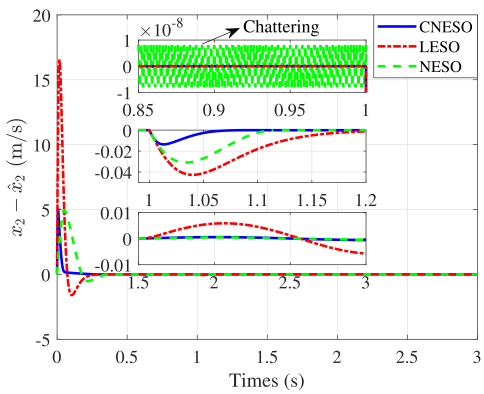

Fig. 4. The estimation error signals of $x_2(t)$ for different ESOs.

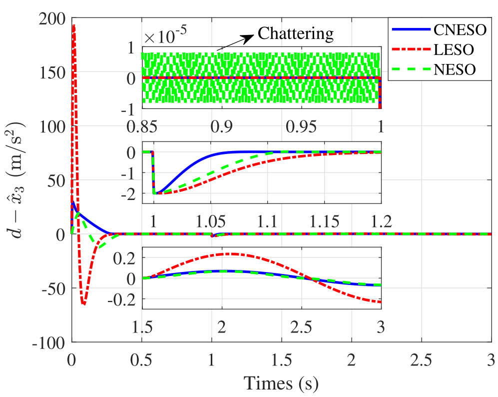

Fig. 5. The estimation error signals of $x_3(t)$ for different ESOs.

TABLE I

QUANTIFIED RESULTS OF DISTURBANCE ESTIMATION ERRORS

$x_1 - \hat{x}_1$	CNESO	LESO	NESO
Peak value (m)	**8.844e-3**	0.110	8.062e-2
Transient time ($\Delta < 5\%$) (s)	**0.051**	0.171	0.285
Maximum steady-state error (m)	**≈0**	4.865e-5	**≈0**
$x_2 - \hat{x}_2$	CNESO	LESO	NESO
Peak value (m/s)	5.053	16.528	**4.956**
Transient time ($\Delta < 5\%$) (s)	**0.047**	0.152	0.287
Maximum steady-state error (m/s)	5.512e-4	5.84e-3	**4.61e-4**
$d - \hat{x}_3$	CNESO	LESO	NESO
Peak value (m/s^2)	29.776	193.132	**19.082**
Transient time ($\Delta < 5\%$) (s)	**0.056**	0.155	0.098
Maximum steady-state error (m/s^2)	6.826e-2	0.2338	**6.586e-2**

$\ell_3 = 2.5$ and $\alpha = 0.2$. The initial values of the system are $x_1(0) = -0.5$, $x_2(0) = 0$. The state and disturbance estimation results are shown in Figs. 3–5 and Table I.

TABLE II

PARAMETERS OF CONTROLLER IN DIFFERENT AXIS

	M_2	L_2	M_1	L_1	k_1	k_2	k_3	k_4
x-axis	2.53	2.53	0.63	0.63	2	4.47	5	0.4
y-axis	2.45	2.45	0.61	0.61	2	4.47	5	0.4
z-axis	4.9	4.9	1.23	1.23	2	4.47	1	0.6

In Table I, the peak value and transient time reflect the transient performance at 0–1.5s, and the maximum steady-state error reflects the steady-state performance at 1.5–3s. It can be seen from Figs. 3–5 and Table I that the CNESO proposed in this paper outperforms LESO and NESO proposed in [27] for transient performance, and also guarantees satisfactory steady-state performance. Moreover, it is observed that the undesired chattering phenomenon by using NESO is successfully eliminated by applying the proposed CNESO.

B. Effectiveness of the Proposed Position Controller for Quadrotor

In the previous subsection, the effectiveness of CNESO is illustrated. Now, we apply the proposed saturated position controller to the quadrotor model established by Simulink, whose parameters are set as $m = 0.7$kg, $J_x = 2.6 \times 10^{-3}$kg $\cdot$ m^2, $J_y = 2.7 \times 10^{-3}$kg $\cdot$ m^2 and $J_z = 4.7 \times 10^{-3}$kg $\cdot$ m^2. The CNESO (13)–(15) and saturated controller (22) are applied here. All the initial states of quadrotor are set to be zero, and a desired trajectory is given as follows [43]

$$p_d = R_x\left(-\frac{\pi}{4}\right) R_z\left(-\frac{\pi}{6}\right) \begin{bmatrix} \frac{\sin(\Phi(t))\cos(\Phi(t))}{\sin^2(\Phi(t))+1} \\ \frac{\cos(\Phi(t))}{\sin^2(\Phi(t))+1} \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

where $R_x(\cdot)$ and $R_z(\cdot)$ are angular rotation matrices defined in [38], and $\Phi(t)$ satisfies $\dot{\Phi}(t) = V\sqrt{1 + \sin^2 t}$. The desired trajectory starts at point $p_{d0} = [0.5,\ -0.61,\ 0.39]^T$, and the desired speed is set to be $V = 0.5$ m/s.

The external disturbances (unit: N) are assumed to be

$$d_1(t) = \begin{bmatrix} 0.2\cos(2\pi t + 5) + 0.3\cos(0.09\pi t + 1) \\ 0.2\sin(0.9\pi t - 2.5) + 0.3\sin(0.25\pi t) \\ 0.2\sin(1.5\pi t) + 0.3\cos(0.75\pi t + 2) \end{bmatrix},$$

$$d_2(t) = [1.7\sin(t + 2), 1.3\cos(1.5t - 1), 0.2\sin(0.5t + 3)]^T .$$

Here, the constraints are given by $T_m = 1.5$mg $= 10.29$N and $\phi_m = \theta_m = 15°$, then the bounds of virtual controller are set to be $u_{m1} = g\sin\theta_m \approx 2.53$, $u_{m2} = g\cos\theta_m \sin\phi_m \approx 2.45$ and $u_{m3} = \frac{T_m}{m} - g \approx 4.9$. The parameters of CNESO are set to be $\varepsilon = 0.05$, $l_0 = 2$, $l_1 = 1.5$, $l_2 = 0.75$, $l_3 = 0.125$, $\ell_1 = 100$, $\ell_2 = 10$, $\ell_3 = 5$, $\alpha = 0.2$, and the parameters of controller are listed in Table II.

With the given parameters, we obtain the trajectory tracking simulation results shown in Figs. 6–8.

On initial time, the nonzero position error leads to the virtual control input $u_i(t)$ saturated immediately which is shown in Fig. 7. It guarantees that the values of $\phi_d(t)$, $\theta_d(t)$ and T are strictly bounded by the desired bounds ϕ_m, θ_m and T_m, respectively, as presented in Fig. 8. A comparative simulation result of position tracking error is shown in Fig. 9,

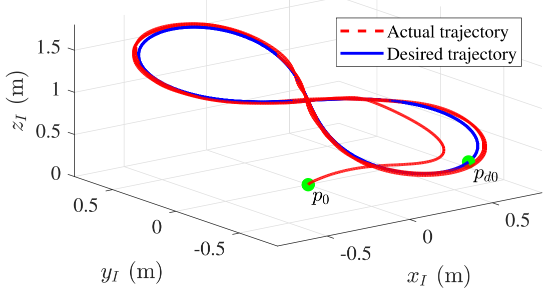

Fig. 6. The quadrotor trajectory in 3D frame.

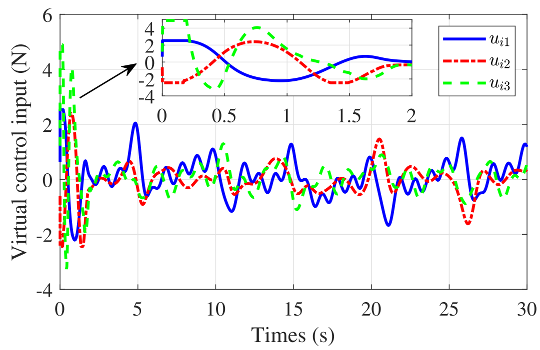

Fig. 7. The virtual control input $u_i(t)$.

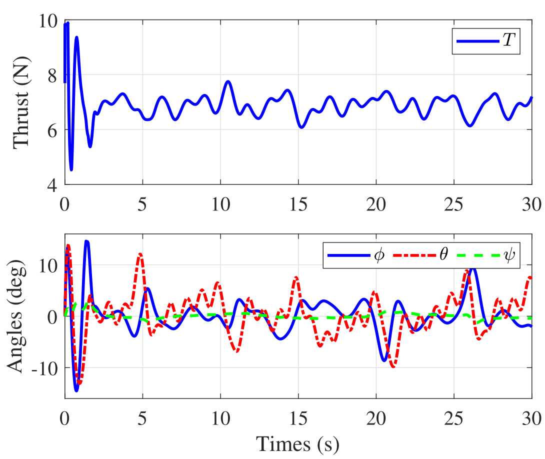

Fig. 8. The total thrust and attitude of the quadrotor.

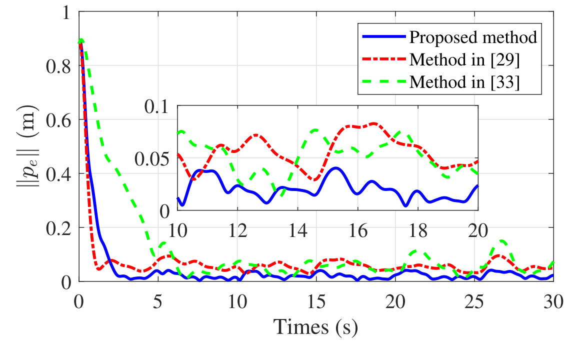

Fig. 9. The comparison of position tracking error with different methods.

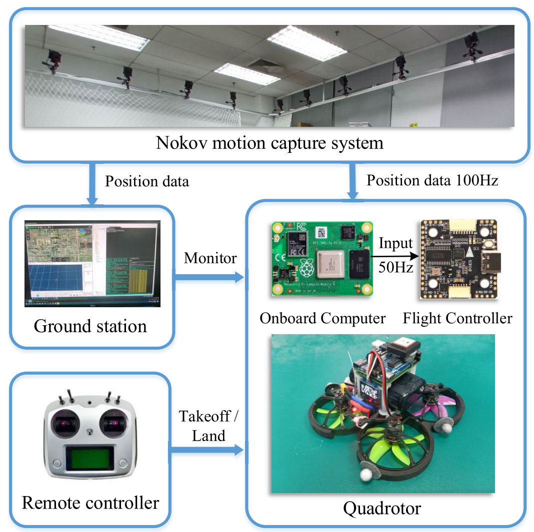

Fig. 10. Framework of quadrotor experimental platform.

where the backpropagating constraints-based control method in [29] and the nested saturation control method without disturbance rejection in [33] are used for comparison. Based on Fig. 9, the mean tracking errors of three different methods in 5-30s are calculated as 0.0208m, 0.0569m and 0.0581m respectively. It can be easily seen that our proposed method performs a higher tracking precision, which further validates the effectiveness of the proposed method.

The simulation results indicate that, by applying the proposed CNESO-based saturated control approach to the quadrotor model with actuator saturation and unknown external disturbances, the desired trajectory tracking objective can be achieved and the control precision is improved greatly.

C. Application of the Proposed Method in Real Flight Experiment

To show the practicability of the proposed method in real quadrotor, a flight experiment is designed in this subsection. The flight controller of quadrotor is kakute h7 mini, which includes STM32H743 processor, BMI270 IMU module and PX4-based firmware. A Raspberry Pi Compute Module 4 is used as the onboard computer, which has quad-core Cortex-A72 @ 1.5GHz CPU, 4GB RAM, 32GB hard drive capacity, and ubuntu 20.04 operating system. The proposed control method runs on the onboard computer, where the control signal is computed in ROS, and the calculated control input is sent to the flight controller at a frequency of 50Hz via a serial portal. The actuator of quadrotor includes 1404 6000KV brushless motor, 4in1 40A ESC and Gemfan D63-5 propellers. A 4000mAh 2s 20C Lipo battery is used for power supply. The mass of quadrotor is $m = 0.325$kg, and the position of quadrotor is obtained by motion capture system, which sends out the measurements to the quadrotor and ground station at a frequency of 100Hz. Besides, the takeoff, landing and mode switching of quadrotor are achieved by remote control, and the states of quadrotor are monitored by ground station in real time. The whole framework of experimental platform is shown in Fig. 10.

A trajectory tracking experiment is designed here, whose goal is to drive the quadrotor to track a circular trajectory:

$$p_d = [\cos(\frac{\pi}{10}t),\ \sin(\frac{\pi}{10}t),\ 0.8]^T,$$

and the initial position of quadrotor is $p_0 = [1,\ 0,\ 0.8]^T$. A PID controller is also used here for comparison, where the parameters P and D are same with k_1 and k_2 and parameter I

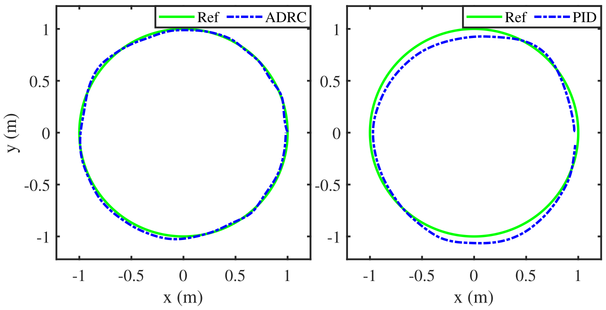

Fig. 11. Trajectory tracking results of different methods in x-y axis.

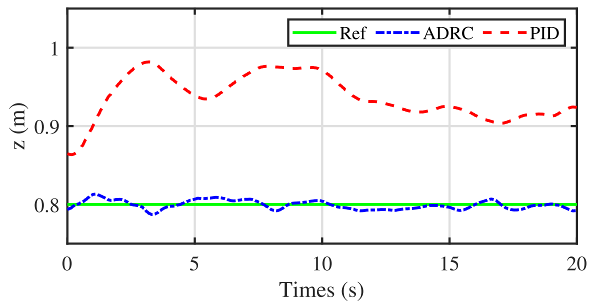

Fig. 12. Trajectory tracking results of different methods in z axis.

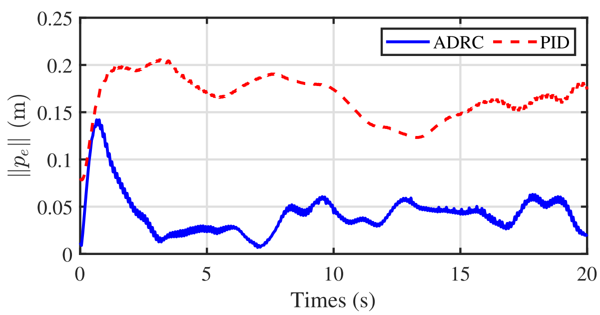

Fig. 13. The comparison of position tracking error with different methods.

is set to 0.3. The tracking results in different axis are shown in Figs. 11–12, where the Ref, ADRC and PID lines respectively represent the reference trajectory, the use of proposed method and PID control method. To better show the difference in tracking error, the comparison result of $\|p_e\|$ is shown in Fig. 13, where the mean tracking errors of proposed method and PID control method are 0.0429m and 0.1655m, respectively. It is clear that the ADRC-based saturated control method performs a higher tracking precision than traditional PID control method, which proves the practicability and efficiency of the proposed method.

VI. Conclusion

In this paper, the trajectory tracking problem of quadrotor with disturbances and input constraints has been investigated. A novel CNESO is developed to estimate the states and lumped disturbances, and the transient performance of the estimation error system can be improved. Moreover, the disturbance estimates are incorporated in the nested saturated position controller to achieve the active disturbance rejection. Simulation and experiment are carried out to verify the effectiveness of the proposed control strategy.

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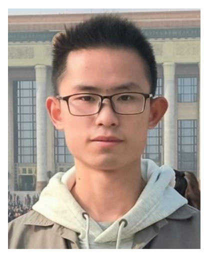

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